

June 17, 2026

To Whom It May Concern,

I am writing to express my strong interest in pursuing a PhD at the University of Southampton, located in Southampton, United Kingdom. The title of this esteemed program is "High Resolution Sensing and Computing for Extreme Conditions". I was particularly drawn to this project due to its focus on high-resolution sensing capabilities for aerospace applications.

My background in MEMS design and signal processing will enable me to contribute meaningfully to the development of lightweight sensing technologies for next-generation flight systems. In my previous work, I have made significant progress in designing efficient state-space models for real-time inference, as seen in my "Medical_{MambaTS}" project. *This linear – time model achieved sub – millisecond bedside inference using PyTorch Autograd.*

I believe that the skills and knowledge gained through this project can be directly applied to the requirements of this PhD offer. Specifically, my expertise in developing low-latency models can be leveraged to create high-resolution sensing technologies for extreme conditions. I am confident that my passion for innovation, combined with my research experience, will enable me to make a valuable contribution to this project.

I am impressed by the University of Southampton's commitment to advancing quantum technology and its applications in aerospace. I am excited about the opportunity to be part of a team that is pushing the boundaries of what is possible in high-resolution sensing and computing for extreme conditions.

I look forward to the opportunity to discuss my application with you further.

Sincerely, Ismail Aissaoui